

Seventh day of class - 9/16

Fun fact: Suppose that $C : ax^2 + bxy + cy^2 + dyz + ez^2 + f = 0$ is a conic. If this conic has real points but no rational points, then there are a finite and *even* number of primes that can be used (as moduli for modular arithmetic) to show there are no rational points on C .

(If the conic has no real points, there will be an odd number of primes.)

Example: $x^2 + 2y^2 = 5$.

This conic has real points. It does not have rational points, and the only primes that can be used to prove it are 2 and 5.

In contrast, for $x^2 + 2y^2 = 6$ there are no primes that can be used to prove it has no points and in fact, this has the point $(2, 1)$ on it.

Last time: We talked about intersection multiplicities, singular points, rational maps and morphisms.

Today: A bit more about morphisms, and transforming equations of curves.

Definition. Two curves C_1 and C_2 are said to be isomorphic if there is a morphism $\phi : C_1 \rightarrow C_2$ that has an inverse $\phi^{-1} : C_2 \rightarrow C_1$ which is also a morphism.

Ex: Let C_1 be the curve with equations $y^2 - 3x^2 = w^2$ and $2z^2 - y^2 = w^2$, and let $E : y^2 = x^3 - 36x$. Then C_1 is isomorphic to E .

Define $\phi : C_1 \rightarrow E$ by

$$\phi(x : y : z : w) = (3y - 3w : -18x : (-1/2)y + z - (1/2)w).$$

The inverse of ϕ is

$$\phi^{-1}(x : y : z) = (-4yz : x^2 + 12xz - 36z^2 : x^2 + 36z^2 : x^2 - 12xz - 36z^2).$$

(Technically, the formulas I wrote for ϕ and ϕ^{-1} are both rational maps, but these can be extended to morphisms by the theorem from the previous class.)

The conclusion is that the map between the rational points on C_1 and the rational points on E

$$\phi : C_1(\mathbb{Q}) \rightarrow E(\mathbb{Q})$$

is a bijection.

Q: How do we come up with these maps?

Section 1.3

How do we transform a cubic curve C with a rational point P on it into $y^2 = x^3 + Ax + B$?

General strategy:

1. Let $Z = 0$ be the equation of the tangent line to C at P .

The multiplicity of the intersection of the tangent line with C at P is either 2, in which case the tangent line intersects the curve in a second point Q , or it is 3, in which case there is no other such point.

2. If the multiplicity is 2, let $X = 0$ be the tangent line through Q . If the multiplicity is 3, let $X = 0$ be any line that doesn't have P on it.
3. Let $Y = 0$ be any line (other than $Z = 0$) that goes through P .
4. Setting $x_1 = X/Z$ and $y_1 = Y/Z$ we get an equation of the form

$$x_1 y_1^2 + (ax + b)y_1 = cx_1^2 + dx_1 + e.$$

We multiply through by x and get

$$x_1^2 y_1^2 + (ax_1 + b)x_1 y_1 = cx_1^3 + dx_1^2 + ex_1.$$

We set $y_2 = x_1 y_1$ and get

$$y_2^2 + (ax_1 + b)y_2 = cx_1^3 + dx_1^2 + ex_1.$$

We then complete the square in y_2 to get

$$y_2^2 + (ax_1 + b)y_2 + \frac{(ax_1 + b)^2}{4} = cx_1^3 + dx_1^2 + ex_1 + \frac{(ax_1 + b)^2}{4}$$

and we set $y_3 = y_2 + \frac{ax_1 + b}{2}$. We now have $y_3^2 = cx_1^3 + fx_1^2 + gx_1 + h$.

If we multiply through by c^2 , we get $c^2 y_3^2 = c^3 x_1^3 + fx_1^2 + gx_1 + h$. We set $y = cy_3$ and $x_2 = cx_1$ we get

$$y^2 = x_2^3 + \frac{f}{c^2}x_2^2 + \frac{g}{c}x_2 + h.$$

Now, the cubic $x_2^3 + (f/c^2)x_2^2 + \frac{g}{c}x_2 + h$ has three roots (over $\mathbb{C}$), and the sum of the roots is $-f/c^2$.

By setting $x = x_2 - \frac{f}{3c^2}$, the roots (in x) are now larger by $f/3c^2$, and so the sum of the roots of the polynomial in x is zero. This gives us an equation of the form

$$y^2 = x^3 + Ax + B.$$

Example: Let $C : a^3 + b^3 + c^3 - 3ab^2 = 0$. This curve has the point $P = (1 : 1 : 1)$ on it. Let $f(a, b, c) = a^3 + b^3 + c^3 - 3ab^2 = 0$.

1. The tangent line to the curve at P has the equation

$$\frac{\partial f}{\partial a}(1, 1, 1)a + \frac{\partial f}{\partial b}(1, 1, 1)b + \frac{\partial f}{\partial c}(1, 1, 1)c = 0.$$

We have $f_a = 3a^2 - 3b^2$, $f_b = 3b^2 - 6ab$, and $f_c = 3c^2$. We get $f_a(1, 1, 1) = 0$, $f_b(1, 1, 1) = -3$ and $f_c(1, 1, 1) = 3$. Thus, the tangent line is $0a - 3b + 3c = 0$ or $c = b$.

We set $Z = c - b$.

2. Plugging this into the curve we find that $f(a, b, c) = f(a, b, b) = a^3 - 3ab^2 + 2b^3 = (a - b)^2(a + 2b)$. Hence, this line intersects the curve in two points: P , and $Q = (-2b : b : b) = (-2 : 1 : 1)$.

The tangent line at Q can be computed in the same way. We get $f_a(-2, 1, 1) = 9$, $f_b(-2, 1, 1) = 15$ and $f_c(-2, 1, 1) = 3$. We get the tangent line to be $9a + 15b + 3c = 0$ or $c = -3a - 5b$.

We set $X = 3a + 5b + c$.

3. The point $(1 : 1 : 1)$ is also on the line $2c = a + b$, so we set $Y = a + b - 2c$.

To transform the equation of the curve it will be helpful to invert this map. We have

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 3 & 5 & 1 \\ 1 & 1 & -2 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

The inverse of the matrix has 9's in the denominators:

$$\begin{bmatrix} 3 & 5 & 1 \\ 1 & 1 & -2 \\ 0 & 1 & -1 \end{bmatrix}^{-1} = \begin{bmatrix} 1/9 & 2/3 & -11/9 \\ 1/9 & -1/3 & 7/9 \\ 1/9 & -1/3 & -2/9 \end{bmatrix}.$$

Clearing those gives

$$9a = X + 6Y - 11Z$$

$$9b = X - 3Y + 7Z$$

$$9c = X - 3Y - 2Z.$$

We now plug these into the cubic, and get

$$\begin{aligned} & f(X + 6Y - 11Z, X - 3Y + 7Z, X - 3Y - 2Z) \\ &= 243(-1/9X^2Z + XY^2 - (10/3)XYZ + 31/9XZ^2 - (7/3)YZ^2 + 23/9Z^3). \end{aligned}$$

We set $x = X/Z$ and $y = Y/Z$ to make this affine and we get the equation

$$xy^2 - \frac{10}{3}xy - \frac{7}{3}y = \frac{1}{9}x^2 - \frac{31}{9}x - \frac{23}{9}$$

Setting $y = xy$ we now get

$$y^2 + (-10/3x - 7/3)y = \frac{1}{9}x^3 - \frac{31}{9}x^2 - \frac{23}{9}x.$$

Completing the square gives

$$y^2 + (-10/3x - 7/3)y + \frac{(10/3x + 7/3)^2}{4} = \frac{1}{9}x^3 - \frac{2}{3}x^2 + \frac{4}{3}x + \frac{49}{36}.$$

The left hand side is $(y - \frac{10x}{6} - \frac{7}{6})^2$ and we set $y = y - (10/3)x + 7/3$. We get

$$y^2 = \frac{1}{9}x^3 - \frac{2}{3}x^2 + \frac{4}{3}x + \frac{49}{36}.$$

Now, we multiply through by 9 and get $9y^2 = x^3 - 6x^2 + 12x + 49/4$. We set $y = 3y$ and have $y^2 = x^3 - 6x^2 + 12x + 49/4$.

To make the equation integral, we can multiply through by 2^6 . We get

$$64y^2 = 64x^3 - 384x^2 + 768x + 784.$$

Now we set $y = 8y$ and $x = 4x$ and we get

$$y^2 = x^3 - 24x^2 + 192x + 784.$$

Finally, to make the x^2 coefficient vanish, we replace x by $x + 8$ and get

$$y^2 = x^3 + 1296.$$

This is quite tedious to work through, and for that reason, we will be using Magma to come up with these transformations.

Section 1.4 - Explicit formulas for the group law

You folks have already worked a bit with this, but I want to talk about it a bit and give some perspective on it.

Let $E : y^2 = x^3 + ax^2 + bx + c$. Saying that E is non-singular is the same as saying that the cubic on the right hand side does not have any repeated roots.

If $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$, then the line through P_1 and P_2 has slope $\lambda = \frac{y_2 - y_1}{x_2 - x_1}$. The y -intercept $\nu = y_1 - \lambda x_1$.

The x -coordinate of $P_1 * P_2$ satisfies the equation

$$(\lambda x + \nu)^2 = x^3 + ax^2 + bx + c.$$

This becomes $x^3 + (a - \lambda^2)x^2 + \dots$. We know that $(x - x_1)$ and $(x - x_2)$ are factors. If $P_1 * P_2 = (x_3, y_3)$, then $(x - x_3)$ is a factor also.

Thus, $a - \lambda^2 = -x_1 - x_2 - x_3$ and so $x_3 = \lambda^2 - a - x_1 - x_2$, and $y_3 = \lambda x_3 + \nu$, and so

$$P_1 + P_2 = (\lambda^2 - a - x_1 - x_2, -\lambda x_3 - \nu).$$

What if $P_1 = P_2$? Then we should use the tangent line to $y^2 = f(x)$. We get $2y \frac{dy}{dx} = f'(x)$ and so $\frac{dy}{dx} = \frac{f'(x)}{2y}$. It turns out in this case that if $P_1 = (x, y)$, the x -coordinate of $2P_1$ is

$$\frac{x^4 - 2bx^2 - 8cx + b^2 - 4ac}{4x^3 + 4ax^2 + 4bx + 4c}$$

provided $4x^3 + 4ax^2 + 4bx + 4c \neq 0$.

Example (if time): Let $E : y^2 = x^3 + x + 1$.

(a) Let $P = (0, 1)$ on E . What is $2P$?

(b) What is $3P = P + 2P$?